

Spring Boot

Simplified creation and execution of production-grade Spring applications.

What it is

Spring Boot makes it easier to create standalone Spring applications with opinionated configuration, starters, embedded servers and production features such as health checks and metrics.

Official site and documentation

Use the official Spring Boot source as the primary reference: <https://spring.io/projects/spring-boot>. Official documentation should be the first source for installation, supported versions, syntax changes and new features.

Prerequisites

Install a JDK compatible with the project version. Maven or Gradle can be used; projects can be generated with Spring Initializr.

Create a first project

Open Spring Initializr, choose Java, Maven or Gradle, an appropriate stable version, Group/Artifact and dependencies such as Spring Web. Generate and extract the project.

Run it

Open it in the IDE and run the main class or use the project Maven/Gradle wrapper. Check port, startup logs and your endpoint.

QA use

QA can run services locally, validate APIs, profiles, variables, databases, health endpoints, logs and integration behavior before E2E automation.

Troubleshooting

Common issues include incompatible JDK, occupied port, wrong profile, missing variables, unavailable database or unresolved dependencies. Read the first meaningful log error.

Installation and configuration - overview

Always use the official website and documentation. Check prerequisites and compatible versions, install for your operating system, configure environment variables or PATH only when required, validate the installation with the recommended command or screen, and run a first project/test before integrating it into the QA workflow.

How to start safely

Start from the official Spring Boot site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Spring Boot, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Using it in a real project

Define where Spring Boot fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

Adoption good practices

Standardize versions and configuration, document important decisions, keep secrets out of code and preserve reproducible examples. Reassess whether Spring Boot still delivers value as the product evolves.

Common issues and troubleshooting

When Spring Boot fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.